Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/30/2024
PI Name:	Yang, Jiseon Ph.D.		
Project Title:	Microbial Social Behavior and Heritable Genetic or Epigenetic Changes Affected by the Spaceflight Environment: Understanding the Evolution of Microbial Interactions During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	11/01/2019	**End Date:**	10/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2025 per NSSC information (Ed., 9/24/24) NOTE: End date changed to 10/31/2024 per NSSC information (Ed., 9/15/23) NOTE: End date changed to 10/31/2023 per F. Hernandez/ARC (Ed., 2/27/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1597		
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Task Description:	September 2023 Update: The Principal Investigator (PI) received official approval from Space Biology on May 30, 2023, to transition the Micro-17 experiment into a ground study (Ed., 9/15/23). Microbes naturally exist as mixed-species communities where they actively communicate and interact with neighboring cells. They adapt and evolve over time within the mixed-species populations. While many studies have shown that the spaceflight environment (and ground-based spaceflight analogs) impacts microbial biological processes, most of the previous experiments were based on single-species cultures. It is still unclear how the spaceflight environment influences microbial social behaviors and how their interaction in the mixed-species community contributes to microbial evolutionary processes in the spaceflight environment. Understanding the effects of the spaceflight environment on the changes in microbial social behavior and the associating evolutionary mechanisms will be necessary for spaceflight design, operations, and astronaut health. It is crucial in regard to preparation for long-duration life support systems during future exploration missions far from Earth. Microbes within the ISS are continuously exposed and adapted to the unique spaceflight environment, which includes microgravity, low fluid shear, and enhanced radiation. A variety of microbial isolates have been recovered from the ISS, including the potable water system. The ISS Water Recovery System recycles urine, condensation, and wastewater. Astronauts rely on the recycled potable water for survival. Most of the microbes recovered from ISS portable water are known biofilm formers, which can increase risks for crew infections or damage to life support system function and integrity. Therefore, it is critical to understand how microbes in the ISS water system adapt and evolve over time. The purpose of the study is to understand the microbial evolution associated with interspecies interaction and to examine the influence of the spaceflight environment on microbial interactions and evolutions in the mixed-species population, by cohort profiling. The hypothesis is that changes in microbial interaction between species of ISS water isolates are associated with microbial genomic evolution during spaceflight, and spaceflight is a contributing factor for the evolution of microbial interaction between different species.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is searching for an understanding by which microorganisms interact and evolve in the context of the complex microbial ecosystem. While immediately applicable to the unique spaceflight environment, it will also be applicable to closed environments in which water can be recycled and reused in terrestrial environments. Overall, this contributes to the creation of healthier and sustainable human built and living environments both in space and on Earth. The study uses the International Space Station as a microbial observatory to study the complex microbial ecosystems and dynamics in human built and living environments, including spaceflights. The study aligns with the research goals in the Space Biology Microbiology Element to determine the effect of spaceflight on life, process, and community dynamics. Completion of this project would 1) provide genomic evidence on the potential microevolution of ISS potable water, 2) provide insight into how microorganisms in the heterogeneous population evolve over time, and 3) determine whether the spaceflight environment is a key contributing factor that influences changes in microbial interaction. The knowledge gained from this study may a) fill knowledge gaps in microbial interspecies dynamics and evolution within the mixed-species population; b) provide novel insight into how the microbial ecological success within the mixed-species population contributes to microbial evolution in human Built Environment, whether in spaceflight or on Earth; and c) provide useful knowledge to support long-duration life support system, particularly, a water recovery system in spaceflight.
Task Progress:	This report provides a summary of the period from September 2023 to August 2024. We have made significant progress on each of the proposed aims since receiving official approval from Space Biology on May 30, 2023, to transition the Micro-17 experiment into a ground study, demonstrating our exceptional expertise and outstanding performance. The project is progressing exceptionally well, with all objectives on track for successful completion. Aim 1: We have successfully collected the necessary data for sequencing and interaction assays. The project continues to advance, with additional analyses and experiments currently in progress. We have sequenced the whole genomes of seven Sphingomonas isolates from the International Space Station (ISS) water system and constructed de novo assemblies for all seven ISS samples. These genomes have been analyzed and compared to seven other ISS or ground isolates. Furthermore, we completed data collection from pairwise comparisons between selected species of ISS isolates to examine microbial interspecies interactions and the changes over time. Aim 2 and Aim 3: We have successfully achieved the milestone of completing the 100-day long-duration culture of mixed species isolated from the ISS or ground environments. DNA and RNA samples were collected at various time points and are now ready for processing. Live bacterial samples have been preserved for future analysis, and planktonic bacterial samples were regularly monitored for contamination and population dynamics. The collected data is currently being analyzed, focusing on population trends in long-term co-culture of ISS water isolates and ground-water isolates. We conducted a comprehensive statistical analysis, including normality tests, regression analysis, nonparametric Wilcoxon paired tests, ANOVA, and REML mixed-effects longitudinal analysis, and identified a significant breakpoint. Additionally, LOWESS (locally weighted scatterplot smoothing) was employed to address nonlinear relationships between groups in the time-series analysis. Moving forward, we will conduct further analysis to accomplish our proposed aims and achieve our goals. Also, we will assess the need for an additional round of 100-day experiments and proceed with execution if deemed necessary.
Bibliography Type:	Description: (Last Updated: 09/02/2025)
Articles in Peer-reviewed Journals	Koroli S, Buss K, Blain JM, Nakka GS, Hong M, McLean RJ, Plugge CM, Yang J. "Draft genome sequence of Sphingomonas paucimobilis strain Sph5, isolated from tap water filtration membrane." Microbiol Resour Announc. 2024 Jan 17;13(1):e0034523. Epub 2023 Dec 1. https://doi.org/10.1128/MRA.00345-23 ; PMID: 38038463; PMCID: PMC10793275. , Jan-2024

Articles in Peer-reviewed Journals	Nickerson CA, McLean RJC, Barrila J, Yang J, Thornhill SG, Banken LL, Porterfield DM, Poste G, Pellis NR, Ott CM. "Microbiology of human spaceflight: Microbial responses to mechanical forces that impact health and habitat sustainability." Microbiol Mol Biol Rev. 2024 Aug 19:e0014423. Epub ahead of print. https://doi.org/10.1128/mmbr.00144-23 ; PMID: 39158275. , Aug-2024